Code No. 8412

OSMANIA UNIVERSITY
FACULTY OF ENGINEERING
UNIVERSITY COLLEGE OF ENGINEERING (AUTONOMOUS)
B.E. (AI&ML) IV-Semester (MAIN) Examinations July-2025

ENGINEERING MATHEMATICS –IV
(Numerical Methods)

100523729008

Time : 3 hours **Max. Marks : 60**

Note : *i) Answer **Question No. 1** (Compulsory) and answer any **four questions** from the remaining questions (2- 7). Each question caries 12 marks.*
ii) Answers must be written in same order as they occur in the Question Paper.
*iii) **Missing data, if any may suitably be assumed.***

		Marks	BT	CO
1. a)	Obtain a partial differential equation by eliminating the arbitrary function from $x+y+z=f(x^2+y^2+z^2)$	2	4	1
b)	Classify the partial differential equation $y\frac{\partial^2 u}{\partial x^2}+2x\frac{\partial^2 u}{\partial x\partial y}+y\frac{\partial^2 u}{\partial y^2}=0$	2	3	2
c)	Construct Newton's divided difference table for the following data.	2	6	3

X:	-3	-2	-1	1	2	3
Y:	18	12	8	6	8	12

d)	Evaluate $\int_0^{10}\frac{dx}{1+x^2}$, by using Simpson's $-\frac{1}{3}$ rule, with width $h=1$.	2	4	4
e)	Explain Taylor series method.	2	3	5
f)	Perform two iterations of the Newton-Raphson method to obtain the approximate value of $(17)^{\frac{1}{3}}$ starting with the initial approximation $x_0=2$.	2	5	2
2. a)	Solve $x^2(y-z)p+y^2(z-x)q=z^2(x-y)$	6	1	1
b)	Find the particular integral of $\frac{\partial^2 z}{\partial x^2}-2\frac{\partial^2 z}{\partial x\partial y}=\sin x\cos 2y$	6	4	1
3. a)	Solve the differential equation $\frac{\partial u}{\partial x}=4\frac{\partial u}{\partial y}$, *where* $u(0,y)=8e^{-3y}$ by the method of Separation of Variables	6	2	2

(P.T.O.)

b) Obtain the Fourier series solution of the one dimensional wave equation 6 4 2

$$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}; 0 < x < l, t > 0$$

with the boundary conditions $u(0,t) = 0;\ u(l,t) = 0; t > 0$
with the initial conditions $u(x,0) = f(x),\ \frac{\partial u}{\partial t}(x,0) = g(x)$.

4. a) Using Decomposition method, solve the system of equations (choose $l_{ii} = 1,\ i = 1,2,3$) $x - y + 5z = 5,\ 2x - 3y + z = 0,\ x + 3y + 7z = 11$ 6 5 3

b) Solve the system of equations 6 6 3
$x - y + 5z = 5, 2x - 3y + z = 0,\ x + 3y + 7z = 11.$

by Gauss-Jacobi iteration method

5. a) From the following table of values x and y obtain $\frac{dy}{dx} \times \frac{d^2y}{dx^2}$ for 6 2 4

x = 2.2

x:	1-0	1-2	1-4	1-6	1-8	2.0	2.2
y:	2.7183	3.3201	4.0552	4.9530	6.0496	7.3891	9.0250

b) Find the value of $y(x)$ from the following data at $x = 2.65$ using Newton's interpolation formula 6 2 4

x	-1	0	1	2	3
$y(x)$	-21	6	15	12	3

6. a) Use Taylor's series method to solve the equation $\frac{dy}{dx} = 3x + y^2,\ y(0) = 1$ and find the value of y for x=0.1 correct to four places of decimals 6 3 5

b) Find the value of y for $x = 0.1$ by Picard's method, given that $\frac{dy}{dx} = \frac{y - x}{y + x},\ y(0) = 1.$ 6 4 5

7. a) Find the solution of the one dimensional heat equation $\frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2}$ *where* $u(0,t) = 0, u(l,t) = 0$ *and* $u(x,0) = x \quad (0 < x < l)$ l being the length of the rod. 6 1 1

b) Solve : $(mz - ny)\frac{\partial z}{\partial x} + (nx - lz)\frac{\partial z}{\partial x} = ly - mx$ 6 4 2

****